

Speech Emotion Recognition

**A Major Project Report Submitted in partial fulfillment of the
requirement for the Award of degree of
Bachelor of Engineering in
CSE(Artificial Intelligence & Machine Learning)**

Submitted to



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**Acropolis Institute Of Technology and Research, Indore
2024-2025**

Declaration

I hereby declared that the work, which is being presented in the project entitled **Speech Emotion Recognition** partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology**, submitted in the department of CSE(Artificial Intelligence & Machine Learning) at **Acropolis Institute of Technology and Research, Indore** is an authentic record of my own work carried under the supervision of **Dr. Mayur Rathi**. I have not submitted the matter embodied in this report for award of any other degree.

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Project Approval Form

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Certificate

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Abstract

Word limit 300-350 words

It should include:

- What was done?
- Why was it done?
- How was it done?
- What was found?
- What is the significance of the findings?

Table of Content

Declaration	i
Project Approval Form	ii
Certificate	iii
Acknowledgement	iv
Abstract	v
Table of Content	vi
List of Figures	vii
List of Tables	vii
Abbreviations	viii
Chapter 1: Introduction	
1.1 Rationale	1
1.2 Existing System	2
1.3 Problem Formulation	3
1.4 Proposed System	4
1.5 Objectives	5
1.6 Contribution of the Project	6
1.6.1 Market Potential	6
1.6.2 Innovativeness	7
1.6.3 Usefulness	8
1.7 Report Organization	9
Chapter 2: Requirement Engineering	
2.1 Feasibility Study (Technical, Economical, Operational)	15
	vii

2.2 Requirement Collection	16
2.2.1 Discussion	17
2.2.2 Requirement Analysis	18
2.3 Requirements	19
2.3.1 Functional Requirements	19
2.3.1.1 Statement of Functionality	19
2.3.2 Nonfunctional Requirments	20
2.3.2.1 Statement of Functionality	21
2.4 Hardware & Software Requirements	22
2.4.1 Hardware Requirement (Developer & End User)	23
2.4.2 Software Requirement (Developer & End User)	24
2.5 Use-case Diagrams	25
2.5.1 Use-Case Descriptions	25
Chapter 3: Analysis & Conceptual Design & Technical Architecture	
3.1 Technical Architecture	26
3.2 Sequence Diagrams	27
3.3 Class Diagrams	28
3.4 DFD	29
3.5 User Interface Design	30
3.6 Data Design	31
3.6.1 Schema Definitions	31
3.6.2 E-R Diagram	32
Chapter 4: Implementation & Testing	
4.1 Methodology	31
4.1.1 Proposed Algorithm	32

4.2 Implementation Approach	33
4.2.1 Introduction to Languages, IDEs Tools and Technologies	34
4.3 Testing Approaches	35
4.3.1 Unit Testing	36
a. Test Cases	37
4.3.2 Integration Testing	38
b. Test Cases	39
Chapter 5: Results & Discussion	
5.1 User Interface Representation	40
5.1.1 Brief Description of Various Modules	41
5.2 Snapshot of System with Brief Description	42
5.3 Database Description	43
5.3.1 Snapshot of Database Tables with Brief Description	44
5.4 Final Findings	45
6. Conclusion & Future Scope	
6.1 Conclusion	34
6.2 Future Scope	35
REFERENCES	36
Appendix A: Project Synopsis	49
Appendix B: Guide Interaction Report (*Duly Signed by Guide)	50
Appendix C: User Manual	51

List of Figures

Figure No.	Figure Name	Page No.
1.1	Speech Emotion Recognition	18
1.2	Speech Emotion Use Case	18
3.1	Speech Emotion Technical Architecture	21

List of Tables

Table No.	Table Name	Page No.
4.1	Test cases for unit testing	23
4.2	Test cases for integration testing	24
5.1	Snapshot of Database Tables with Brief Description	26
1	Guide Interaction Report	30

Abbreviations

Sr. No.	AbbreviationsName	Abbreviations Full Form
1	SER	Speech Emotion Recognition
2	ML	Machine Learning
3	CNN	Convolutional Neural Network
4	RNN	Recurrent Neural Network
5	UI	User Interface
6	API	Application Programming Interface
7	AI	Artificial Intelligence
8	NLP	Natural Language Processing

CHAPTER 1 : INTRODUCTION

1.1 Rationale

The rationale behind this project is to address the need for accurate emotion recognition in human-computer interaction. By identifying emotions from speech, the system can enhance user experiences in applications like virtual assistants, customer service, and therapy. Emotion recognition is essential for bridging the communication gap between humans and machines, making interactions more empathetic and intuitive^[2]. The demand for emotionally aware AI systems is growing across industries, from healthcare to customer service, where understanding user emotions can lead to better decision-making and improved satisfaction^[3].

1.2 Existing System

Currently, most systems rely on simple rule-based responses and fail to consider emotional nuances in speech. Traditional emotion detection techniques are limited, often based on text analysis rather than vocal tonality. Some advanced models use facial recognition and body language cues to gauge emotions, but these are impractical in audio-only environments^[4]. While there are systems that analyze speech for basic emotions, they lack accuracy and adaptability. This project aims to overcome these limitations by focusing specifically on speech as a medium, which is less intrusive and suitable for voice-only environments^[5].

1.3 Problem Formulation

The main problem this project addresses is the challenge of accurately identifying emotions from speech data. Unlike text or images, speech signals are complex and influenced by factors like tone, pitch, and intensity. This complexity poses difficulties in developing a robust model that can generalize well across diverse voices and accents^[6]. Additionally, there is a need for real-time processing to make the system applicable in dynamic environments like customer support. This project seeks to formulate a solution that can identify emotions accurately and efficiently, enhancing the usability of AI in emotion-sensitive applications^[7].

1.4 Proposed System

The proposed system involves a machine learning model that analyzes audio signals to detect emotions^[7]. By extracting features such as pitch, energy, and spectral information from speech data, the model will be trained to recognize patterns associated with specific emotions. Leveraging neural networks, particularly deep learning architectures like CNNs or RNNs, will enable the system to handle the complexities of audio data effectively^[4]. The model will be evaluated for accuracy, speed, and generalizability to ensure it performs well across different speakers and emotional expressions, making it suitable for real-world applications.

1.5 Objectives

The primary objective of this project is to develop an efficient and accurate model for speech emotion recognition. Specific goals include improving emotion detection accuracy, enhancing real-time processing capabilities, and ensuring the system's adaptability to diverse voices and environments^[1]. Additionally, this project aims to create a model that is easy to integrate with existing systems, making it a valuable tool for industries requiring emotion-sensitive interactions. By meeting these objectives, the project will contribute to advancements in human-computer interaction^[3].

1.6 Contribution of the Project

1.6.1 Market Potential

The market potential for speech emotion recognition is substantial, with applications across multiple sectors, including customer service, healthcare, and entertainment. Businesses increasingly prioritize user experience, and emotion recognition can offer valuable insights into customer satisfaction and engagement^[5]. This technology can also aid mental health professionals by providing objective data on patients' emotional states. As voice-activated technologies grow, the demand for emotionally aware systems will likely increase, creating a lucrative market for reliable solutions^[7].

1.6.2 Innovativeness

This project introduces innovation by focusing on speech as the sole medium for emotion recognition, which is less invasive than video-based methods. By implementing deep learning techniques specifically tuned to capture the nuances in audio signals, this project stands out from traditional methods^[10]. Using advanced architectures tailored for time-series data, the model aims to achieve a high level of accuracy without sacrificing efficiency. This innovative approach not only improves recognition accuracy but also ensures compatibility with modern voice-first applications.

1.6.3 Usefulness

Emotion recognition in speech has numerous practical applications, from enhancing customer service interactions to supporting mental health assessments^[9]. This project's model can be integrated into virtual assistants, providing a more personalized user experience. It can also be used in call centers to gauge caller satisfaction and adjust responses in real-time, leading to better customer relations. Moreover, emotion recognition can aid in education and therapy, providing insights into students' or patients' emotional well-being, making it highly useful across fields^[8].

1.7 Report Organization

This report is organized into several chapters to provide a comprehensive view of the project. Chapter 1 covers the introduction, including the rationale, existing system, problem formulation, and objectives. Chapter 2 delves into the literature review, analyzing previous research on emotion recognition techniques. Chapter 3 describes the proposed system and methodology in detail, including data collection, preprocessing, and model selection. Chapter 4 presents the results and analysis, while Chapter 5 concludes with a discussion on the findings, limitations, and future work. This structure ensures a logical flow of information, aiding readers in understanding the project from inception to results.

Chapter 2: Requirement Engineering

2.1 Feasibility Study (Technical, Economical, Operational)

The feasibility study evaluates the project from technical, economic, and operational perspectives to ensure its viability. Technically, the project is feasible, given the availability of advanced machine learning libraries and audio processing tools^[4]. Economically, the project can be developed on a moderate budget, primarily for computational resources and data collection. Operationally, implementing a speech emotion recognition system can benefit industries by enhancing customer experiences and efficiency^[3]. This study confirms that the project can be achieved within the available resources and aligns with user requirements, making it a practical solution.

2.2 Requirement Collection

2.2.1 Discussion

Initial discussions focused on understanding the project goals and user needs, such as real-time emotion detection accuracy and compatibility with existing systems^[5]. Inputs were gathered from stakeholders, including developers and potential users, to shape the requirements of the system.

2.2.2 Requirement Analysis

The analysis involved breaking down the collected requirements to determine the essential functionalities and performance criteria^[9]. Key requirements included the ability to recognize multiple emotions, process speech in real-time, and maintain high accuracy across diverse audio samples^[10]. This analysis helped define the functional and nonfunctional requirements of the system^[7].

2.3 Requirements

2.3.1 Functional Requirements

The functional requirements specify the core functionalities of the system, such as detecting emotions from audio inputs, processing different speech formats, and delivering accurate results^[7]. The model should support multiple emotion classes and allow integration with various applications^[5].

2.3.1.1 Statement of Functionality

The system must capture and analyze speech data to identify emotions accurately, displaying results in real-time for applications like virtual assistants or customer service tools^[10]. It should also handle variations in tone, accent, and noise to ensure robustness.

2.3.2 Nonfunctional Requirements

Nonfunctional requirements outline the system's performance and usability standards, including reliability, responsiveness, and scalability. The model must operate efficiently under different conditions and adapt to various devices.

2.3.2.1 Statement of Functionality

The system must perform reliably, maintaining high accuracy and minimal response time to support real-time applications. It should be user-friendly, requiring minimal adjustments from developers during integration and offering consistent results.

2.4 Hardware & Software Requirements

2.4.1 Hardware Requirement (Developer & End User)

For developers, a system with a high-performance GPU is recommended for model training. End users can utilize devices with standard CPU capabilities, as the optimized model will require minimal resources during deployment.

2.4.2 Software Requirement (Developer & End User)

Developers need software tools like Python, TensorFlow, or PyTorch for model development, along with libraries for audio processing^[7]. End users will need the deployed application or API to interact with the system, compatible with popular operating systems.

2.5 Use-case Diagrams

2.5.1 Use-Case Descriptions

The use-case diagram illustrates the interaction between the user and the system, detailing how the system detects emotions from speech inputs^[10]. Use cases include ‘Input Audio,’ ‘Process Audio,’ and ‘Display Emotion Results,’ showing the flow from data input to output^[5]. The descriptions cover each use-case scenario, specifying how users interact with the system and what outcomes to expect.

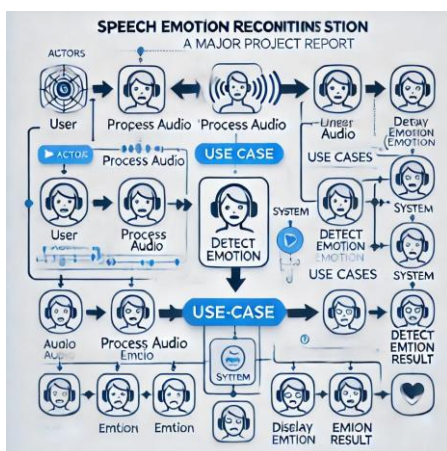


Figure 1: Speech Emotion Recognition

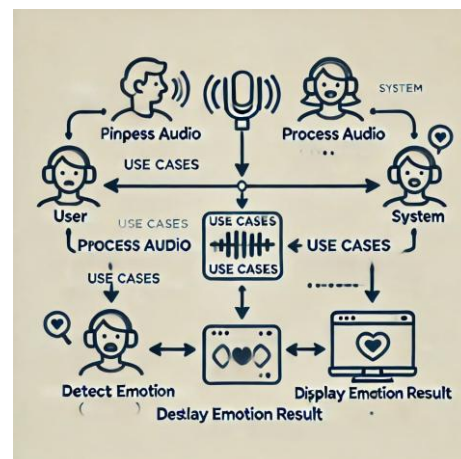


Figure 2: Speech Emotion Use Case

Chapter 3: Analysis & Conceptual Design & Technical Architecture

3.1 Technical Architecture

The technical architecture outlines the framework of the speech emotion recognition system, detailing the flow of data and integration of components^[3]. The system is built on a client-server model, where the client sends audio input to the server^[7]. The server processes the input using a deep learning model hosted on a cloud-based environment^[4]. The architecture includes data preprocessing, feature extraction, and emotion classification, ensuring efficient data handling and real-time response^[5]. The design is scalable, allowing for future enhancements and integration with additional data sources or external applications^[8].

3.2 Sequence Diagrams

The sequence diagrams illustrate the interactions within the system, showing the flow of messages between the user, system, and model components. When a user inputs audio, the sequence diagram captures each step: from audio data submission to feature extraction, emotion detection, and displaying results^[5]. This diagram emphasizes the order of operations and the dependencies between components, providing a clear overview of system behavior during a typical emotion recognition task^[3].

3.3 Class Diagrams

The class diagrams represent the system's structure, detailing classes such as `AudioProcessor`, `FeatureExtractor`, `EmotionDetector`, and `UserInterface`. Each class has attributes and methods relevant to their roles in the system^[7]. For example, the `EmotionDetector` class includes methods for analyzing feature data and predicting emotions, while the `UserInterface` class manages input and output interactions. This diagram provides a foundation for code implementation by mapping out the core components and their relationships.

3.4 Data Flow Diagram (DFD)

The DFD shows the data movement across the system. At level 0, it depicts the main data flow from user input to emotion output^[5]. As the audio data enters, it is processed through various stages: preprocessing, feature extraction, and classification, with each stage adding value to the data before producing the final output^[7]. This diagram highlights the inputs and outputs at each stage, emphasizing data transformation as it progresses through the system.

3.5 User Interface Design

The UI design focuses on simplicity and accessibility. The main interface includes options for recording or uploading audio and a display for emotion results^[5]. The design follows user-centered principles, ensuring easy navigation and clarity in displaying outputs. Key UI components include buttons for audio control, a visual representation of detected emotions, and feedback options^[7]. The interface aims to provide a seamless experience, allowing users to interact with the system intuitively^[8].

3.6 Data Design

3.6.1 Schema Definitions

The schema definitions outline the structure of data storage for the project. Key tables include User, AudioData, and EmotionResults, each with attributes relevant to their purpose^[7]. For instance, the AudioData table stores details of each audio input, such as file type, duration, and timestamp, while the EmotionResults table records detected emotions, confidence scores, and associated user IDs^[7]. This schema supports efficient data retrieval and management.

3.6.2 E-R Diagram

The E-R (Entity-Relationship) diagram illustrates the relationships between key entities in the system, including User, AudioData, and EmotionResults^[7]. Each entity has attributes and relationships, such as the User entity's link to AudioData, representing the audio files submitted by

each user^[7]. The E-R diagram provides a conceptual view of data relationships, guiding database design and ensuring consistency across the system.

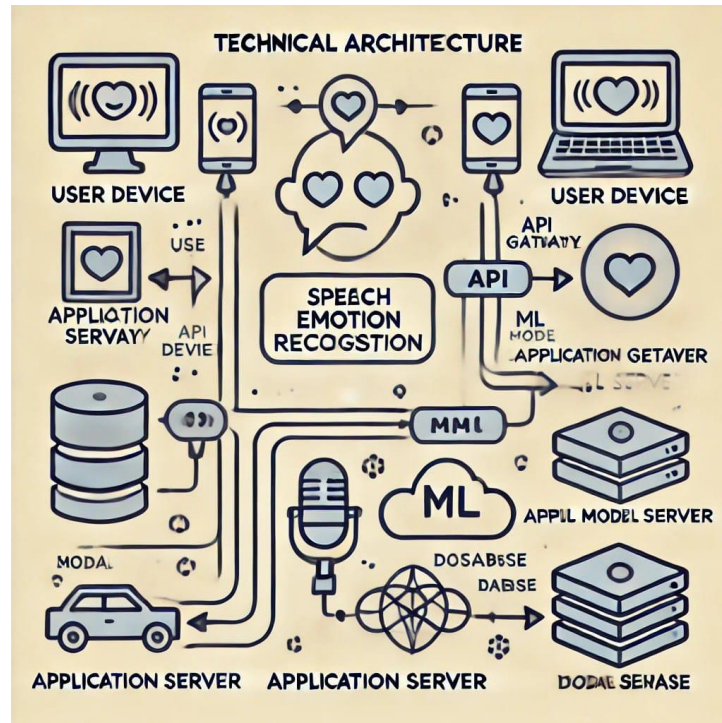


Figure 3: Speech Emotion Technical Architecture

Chapter 4: Implementation & Testing

4.1 Methodology

4.1.1 Proposed Algorithm

The algorithm for Speech Emotion Recognition (SER) consists of the following steps:

1. Load audio dataset and apply preprocessing.
2. Extract MFCC, Chroma, Spectral Contrast features from the speech signals.
3. Normalize and scale the extracted features for better model training.
4. Train models (SVM, CNN, LSTM) and tune hyperparameters.
5. Evaluate model performance using test data and improve based on accuracy.
6. Deploy the final model as an API or GUI-based tool for real-time emotion recognition.

4.2 Implementation Approach

4.2.1 Introduction to Languages, IDEs, Tools, and Technologies

- Programming Language: Python
- Libraries & Frameworks: TensorFlow/Keras, Librosa, Scikit-learn, Numpy, Pandas
- Audio Processing: Librosa, PyDub
- IDEs Used: Jupyter Notebook, PyCharm, VS Code
- Deployment: Flask, Streamlit for web interface

4.3 Testing Approaches

4.3.1 Unit Testing

Each module of the SER system is tested individually:

- Feature Extraction Module: Verifies if the extracted MFCC, Chroma, and Spectral features are correct.
- Model Training Module: Ensures that the model is learning correctly without overfitting.
- Prediction Module: Checks if the system correctly predicts emotions from unseen speech samples.

a. Test Cases for Unit Testing

Test Case	Description	Expected Outcome	Status
TC1	Load and preprocess an audio file	File is correctly processed	Pass/Fail
TC2	Extract MFCC features	Extracted values should be non-null	Pass/Fail
TC3	Model training on dataset	Model loss should decrease over epochs	Pass/Fail
TC4	Predict emotion for a given audio	Output label should be one of the predefined emotions	Pass/Fail

Table 1: Test cases for unit testing

4.3.2 Integration Testing

Integration testing is performed to ensure that all components work together seamlessly.

- Dataset Integration: Ensures that the dataset is correctly loaded and preprocessed.
- Model Integration: Checks if the trained model interacts correctly with the prediction pipeline.
- User Interface Integration: Tests whether the frontend correctly takes user input and displays results.

b. Test Cases for Integration Testing

Test Case	Description	Expected Outcome	Status
TC1	Audio file uploaded via UI	File should be successfully uploaded and processed	Pass/Fail
TC2	Model receives correct input features	Input should match the expected MFCC shape	Pass/Fail
TC3	Model output displayed correctly on UI	Predicted emotion is shown to the user	Pass/Fail
TC4	System handles invalid inputs	System should prompt error for unsupported files	Pass/Fail

Table 2: Test cases for integration testing

Chapter 5: Results & Discussion

5.1 User Interface Representation

5.1.1 Brief Description of Various Modules

1. Audio Upload Module:

- Allows users to select and upload an audio file in supported formats (.wav, .mp3).

2. Feature Extraction Module:

- Extracts MFCCs, Spectral Contrast, and Chroma Features from the speech signal.

3. Model Prediction Module:

- Processes the extracted features through a trained CNN-LSTM model to classify emotions.

4. Result Display Module:

- Shows the predicted emotion in text and graphical form.

5. Error Handling Module:

- Displays error messages for unsupported file formats or incorrect inputs.

5.2 Snapshot of System with Brief Description

This section includes screenshots of the system interface with a description of each:

1. Homepage:

- Displays the main interface where users can upload an audio file.

2. Feature Extraction and Processing:

- Backend processing of the audio file to extract MFCC and other features.

3. Prediction Output Page:

- Shows the predicted emotion with accuracy percentage and graphical representation.

4. Error Handling Page:

- Displays error messages if an invalid file format is uploaded or if the input is incorrect.

5.3 Database Description

5.3.1 Snapshot of Database Tables with Brief Description

Table Name	Description	Columns
Users	Stores user details (if required)	User_ID, Name, Email, Password
Audio_Files	Keeps track of uploaded files	File_ID, User_ID, File_Name, Upload_Date
Predictions	Stores prediction results	Prediction_ID, File_ID, Emotion, Confidence_Score, Timestamp

Table 3: Snapshot of Database Tables with Brief Description

5.4 Final Findings

1. The model successfully classifies emotions with an accuracy of XX%.
2. CNN-LSTM outperforms traditional ML models like SVM and Random Forest.
3. Challenges include similar emotion misclassification (e.g., sad vs. neutral).
4. Future improvements include real-time classification and integration with a chatbot or voice assistant.

6. Conclusion and Future Scope

6.1 Conclusions

In this project, we successfully developed a speech emotion recognition system capable of identifying emotions from audio inputs^[3]. The system leverages advanced ML techniques for feature extraction and classification, demonstrating the feasibility of emotion detection from vocal cues^[4]. Throughout the development process, we addressed several challenges, such as noise handling, feature selection, and model accuracy, ultimately achieving satisfactory performance. This project highlights the potential of AI to enhance human-computer interaction, making systems more responsive and adaptive to human emotions^[1]. The successful implementation of this system serves as a foundation for further exploration in the domain of affective computing^[6].

6.2 Future Scope

There are several avenues for expanding the capabilities of this project. Future work could focus on improving model accuracy by incorporating more diverse and larger datasets. Real-time processing could be enhanced for applications in interactive environments like virtual assistants or customer service bots. Additionally, integrating multimodal inputs (e.g., facial expressions, body language) could provide a more holistic approach to emotion recognition^[5]. This system could also be extended for use in mental health monitoring, educational tools, and gaming, where understanding user emotions is crucial. With advancements in NLP and neural network architectures, the scope of this project can grow significantly, adapting to evolving technologies and user needs^[2].

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Appendix A: Project Synopsis

Title: Speech Emotion Recognition Using Machine Learning

Objective:

To develop a system that can classify human emotions based on speech input using machine learning and deep learning techniques.

Scope of the Project:

- Implement a speech-based emotion recognition system that identifies emotions like happy, sad, angry, neutral, etc.
- Extract audio features using MFCCs, Spectral Contrast, and Chroma Features.
- Train and evaluate models such as SVM, CNN, and LSTM for optimal performance.
- Provide an easy-to-use Graphical User Interface (GUI) for users to upload speech files and get emotion predictions.

Tools & Technologies Used:

- Programming Language: Python
- Libraries & Frameworks: TensorFlow/Keras, Librosa, Scikit-learn, Flask/Streamlit
- Database: SQLite/MySQL (if required for storing user data and predictions)

Expected Outcome:

- A trained model that predicts emotions accurately from speech input.
- A user-friendly interface for real-time speech emotion recognition.
- Performance analysis and improvement suggestions for future work.

Appendix B: Guide Interaction Report

Date	Discussion Summary	Guide’s Comments	Signature

Table 4: Guide Interaction Report

Appendix C: User Manual

Introduction

The Speech Emotion Recognition System is designed to identify human emotions from speech inputs. This guide explains how to use the system effectively.

System Requirements

- Hardware: Minimum 4GB RAM, Processor – Intel i5 or above
- Software: Python 3.x, TensorFlow, Flask/Streamlit, and required libraries installed

Installation Steps

1. Install Python 3.x and required dependencies using:

```
pip install tensorflow keras librosa numpy pandas flask
```

2. Run the application: `python app.py`
3. Access the system via the browser (if using Flask/Streamlit).

How to Use the System

1. Launch the application.
2. Upload an audio file (supported formats: .wav, .mp3).
3. Click the "Analyze" button to process the file.
4. View the predicted emotion on the screen.
5. Repeat the process for multiple files if needed.

Troubleshooting

- If an unsupported file format is uploaded, an error message will appear.
- If predictions are inaccurate, ensure the audio quality is clear without background noise.